

NiCE Search Grid System Requirements Guide Version 20.1.2.1

Introduction

This topic lists the requirements that your system must meet before you begin installing NiCE Search Grid.

Requirements

The minimum and recommended requirements for NiCE Search Grid are displayed in the following table.



Although NiCE Search Grid will run if you adhere to the minimum requirements, you will achieve the best performance if your system meets the recommended requirements.

Component	Minimum	Recommended (if different)
Processor (See important notes below table for further details about Hyper-Threading, NUMA and RAM requirements implications of core count.)	- Intel Core i7, Quad Core, 8 MB L3 cache - No more than 64 logical cores if running under Windows - Hyper-Threading disabled (unless using a GPU - see note below) - Non-uniform memory access (NUMA) disabled	Intel Xeon E5-2690, Octa Core, 20 MB L3 cache
RAM (See important notes below table for other considerations regarding RAM requirements.)	32 GiB for phrase recognition; 8 GiB otherwise	96 GiB for phrase recognition; 16 GiB otherwise

Component	Minimum	Recommended (if different)
Available Disk Space for Installation	2.6 GiB	
Available Disk Space for Data Files	- With large scale search enabled, 3.9 MiB per media hour - With large scale search disabled, 1.4 MiB per media hour - Assumptions - Measurements do not include media file storage - Using latest, compatible language pack version - Average media duration at least five minutes - No long-term storage of phrase recognition results	
GPU (if used)	- If using ONLY pre-v10.8 phrase packs - NVIDIA A10 GPU - Per GPU RAM: 3.2 MiB per phrase pack + (<i>1.5 MiB GPU RAM per CPU core</i> / number of GPUs) - If using ANY v10.8 phrase packs - Dual NVIDIA A10 GPUs - Per-GPU RAM: - v10.8 Small Model: 3.4 GiB	Dual NVIDIA A10 GPUs

Component	Minimum	Recommended (if different)
	per phrase pack + (465 MiB per CPU core / number of GPUs) ▪ v10.8 Large Model: 5.1 GiB per phrase pack + (1.35 GiB per CPU core / number of GPUs) • CUDA version 11.4.0 GA or later • Driver version >=470.72	
Other Hardware	• One 1 Gbps NIC	• Two 1 Gbps NICs • Solid State Drives
Operating Systems	Windows Server <i>with Desktop Experience</i> feature installed: • 2016 • 2019 • 2022	2022
Language Packs	Version 7.0	Version 9.3
Java Runtime Environment	Oracle Server JRE 17, but NiCE <i>strongly</i> recommends using Search Grid's default JRE	Search Grid's default / embedded JRE



The more cores that there are, the more RAM that is required in order to take advantage of the parallel processing those cores offer, so more than the recommended 96 GiB may be required for high-core-count servers. If there is not enough RAM to support the number of cores, and site-specific phrase recognition configuration has overridden the default configuration values, especially if done via the `maxConcurrentOperations` property, the maximum number of concurrent phrase recognition operations may

be too high, potentially causing out-of-memory failures.

It should also be noted that Hyper-Threading effectively doubles the core count from the number of physical cores. Therefore, unless there is sufficient RAM to accommodate the number of logical cores, Hyper-Threading can cause excessive memory use and potential out-of-memory failures, and therefore should be disabled *unless a GPU is being used*. On Compute Nodes, or on Data Nodes that allow phrase recognition operations, *Hyper-Threading should always be **disabled** unless a GPU is being used*, because it can actually degrade performance.

Please see the **Configuring Phrase Recognition** section of the **Search Grid Administration Guide** for further information.

Virtual Memory should not be enabled, or at least it should be configured such that the initial size and maximum size are both set to the minimum value allowed by the operating system (for Windows that is usually something very low). Best is to disable it altogether if possible. Search Grid is simultaneously very CPU-intensive AND very RAM-intensive, so if it were having to swap to/from the virtual memory on disk, its performance would suffer greatly. This is especially true for phrase recognition and search operations.



NUMA is another consideration on servers that have multiple physical CPU sockets/packages. NUMA can cause a significant degradation in performance and should be disabled. **However**, the terminology for the NUMA settings vary by server manufacturer. For example, the proper setting in one BIOS may be to set NUMA to Disabled, while another manufacturer's BIOS may require it to be set to NUMA Interleaving Enabled. Therefore, consulting the manufacturer's configuration guide is essential.